

Fundamentals of Machine Learning

Practice Sheet 2 — Eigendecomposition, the SVD & Low-Rank Approximation

Practice set

This sheet focuses on the structure that makes symmetric matrices, the singular value decomposition and low-rank approximation so useful in machine learning. Work each part by hand and show your steps. Several questions reuse the same small matrix on purpose, so the arithmetic stays light and you can see how the pieces fit together. A handful of $\sqrt{2}$ and $\sqrt{5}$ factors appear — keep them symbolic rather than reaching for a calculator.

Q1. Diagonalisation of a symmetric matrix

Let $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.

- Find the eigenvalues of A and a *unit* eigenvector for each. Assemble them into an orthogonal matrix $Q = [u_1 \ u_2]$ and a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2)$, so that $A = Q\Lambda Q^\top$.
- Verify that $Q^\top Q = I$ (i.e. Q is orthogonal, so $Q^{-1} = Q^\top$), and confirm that $Q\Lambda Q^\top$ reproduces A .
- Using the decomposition $A = Q\Lambda Q^\top$ — and *without* multiplying matrices out — state the eigenvalues of A^2 and of A^{-1} . In one sentence, say why $A^k = Q\Lambda^k Q^\top$ holds for any integer k .

Q2. Spectral decomposition as a sum of rank-one projections

Keep $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and the unit eigenvectors u_1, u_2 from Q1.

- Form the two outer products $u_1 u_1^\top$ and $u_2 u_2^\top$ as 2×2 matrices.
- Verify the spectral decomposition $A = \lambda_1 u_1 u_1^\top + \lambda_2 u_2 u_2^\top$ by adding the two weighted matrices and checking you recover A .
- In two or three sentences, explain what this says geometrically: each term $u_i u_i^\top$ projects a vector onto the i -th eigen-direction, and A acts by scaling those projections by the eigenvalues λ_i .

Q3. Positive definiteness and quadratic forms

Keep $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.

- Show that A is positive definite in two ways: via its eigenvalues, and via its leading principal minors (the top-left entry and $\det A$).
- Evaluate the quadratic form $q(x) = x^\top A x$ at $x = [1, 1]^\top$ and at $x = [1, -1]^\top$. Check that each value equals $\lambda \|x\|_2^2$ for the appropriate eigenvalue, and say why that identity holds when x is an eigenvector.
- Show that for *any* matrix B , the matrix $B^\top B$ is positive semidefinite, by expanding $x^\top B^\top B x$. Connect this to why the normal-equations matrix $X^\top X$ that appears in least-squares regression is always positive semidefinite.

Q4. Singular value decomposition

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

- Compute $A^\top A$, find its eigenvalues, and hence the singular values $\sigma_1 \geq \sigma_2$ of A (the singular values are the square roots of the eigenvalues of $A^\top A$).
- Find the right singular vectors v_1, v_2 (unit eigenvectors of $A^\top A$) and the left singular vectors via $u_i = A v_i / \sigma_i$. Write down U, Σ, V so that $A = U \Sigma V^\top$. (*Hint: A is symmetric, so $A^\top A = A^2$.*)
- This A has a negative eigenvalue, yet all its singular values are positive. Explain in two or three sentences how the SVD relates to the eigendecomposition for a symmetric matrix ($\sigma_i = |\lambda_i|$), and why the SVD exists for *every* matrix while the eigendecomposition does not.

Q5. Low-rank approximation

Return to $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, whose singular values are $\sigma_1 = 4$ and $\sigma_2 = 2$ with singular vectors $u_1 = v_1 = \frac{1}{\sqrt{2}}[1, 1]^\top$ and $u_2 = v_2 = \frac{1}{\sqrt{2}}[1, -1]^\top$ (it is symmetric and positive definite, so its SVD equals its eigende-

composition).

- (a) Write down the best rank-1 approximation $A_1 = \sigma_1 u_1 v_1^\top$ as an explicit 2×2 matrix.
- (b) Compute the approximation error in the Frobenius norm, $\|A - A_1\|_F = \sqrt{\sum_{ij} (A - A_1)_{ij}^2}$, and confirm it equals the discarded singular value σ_2 .
- (c) State the Eckart–Young theorem in one sentence (which rank- k matrix best approximates A , and in what sense), and give one concrete use of low-rank approximation in machine learning or data compression.

Q6. From eigendecomposition to PCA

A small centred dataset has the 2×2 covariance matrix $C = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$.

- (a) Find the eigenvalues and unit eigenvectors of C . (*You should find that $\det C = 0$.*)
- (b) Identify the first principal component (the eigenvector with the largest eigenvalue) and compute the fraction of the total variance it captures, $\lambda_1/(\lambda_1 + \lambda_2)$. What does your answer say about the effective dimensionality of this dataset?
- (c) In two or three sentences, explain how PCA is an eigendecomposition of the covariance matrix, and how keeping only the top eigenvector(s) is exactly a low-rank approximation of the data — tying this question back to Q1 and Q5.